Lemma 1.4. *An absolute value $|\ |$ on a field k is nonarchimedean if and only if*

$$|\underbrace{1+\cdots+1}_{n}| \le 1$$

for all $n \ge 1$.

Proof. See Problem Set 1. □

Corollary 1.5. *In a field of positive characteristic every absolute value is nonarchimedean, and the only absolute value on a finite field is the trivial one.*

Definition 1.6. Two absolute values $|\ |$ and $|\ |'$ on the same field k are *equivalent* if there exists an $\alpha \in \mathbb{R}_{>0}$ for which $|x|' = |x|^\alpha$ for all $x \in k$.

1.3 Absolute values on $\mathbb{Q}$

To avoid confusion we will denote the usual absolute value on $\mathbb{Q}$ (inherited from $\mathbb{R}$) by $|\ |_\infty$; it is an archimedean absolute value. But there are are infinitely many others. Recall that any element of $\mathbb{Q}^\times$ may be written as $\pm\prod_q q^{e_q}$, where the product ranges over primes and the exponents $e_q \in \mathbb{Z}$ are uniquely determined (as is the sign).

Definition 1.7. For a prime p the *p-adic valuation* $v_p\colon \mathbb{Q} \to \mathbb{Z}$ is defined by

$$v_p\left(\pm\prod_q q^{e_q}\right) := e_p,$$

and we define $v_p(0) := \infty$. The *p-adic absolute value* on $\mathbb{Q}$ is defined by

$$|x|_p := p^{-v_p(x)},$$

where $|0|_p = p^{-\infty}$ is understood to be 0.

Theorem 1.8 (Ostrowski's Theorem)**.** *Every nontrivial absolute value on $\mathbb{Q}$ is equivalent to $|\ |_p$ for some $p \le \infty$.*

Proof. See Problem Set 1. □

Theorem 1.9 (Product Formula)**.** *For every $x \in \mathbb{Q}^\times$ we have*

$$\prod_{p\le\infty} |x|_p = 1.$$

Proof. See Problem Set 1. □

1.4 Discrete valuations

Definition 1.10. A *valuation* on a field k is a group homomorphism $k^\times \to \mathbb{R}$ such that for all $x, y \in k$ we have

$$v(x+y) \ge \min\bigl(v(x), v(y)\bigr).$$

We may extend v to a map $k \to \mathbb{R} \cup \{\infty\}$ by defining $v(0) := \infty$. For any any $0 < c < 1$, defining $|x|_v := c^{v(x)}$ yields a nonarchimedean absolute value. The image of v in $\mathbb{R}$ is the